

Enabling Emoji Display for C++ Projects in Visual Studio (Windows)

This guide provides steps to force **UTF-8 encoding** for your C++ console application in Visual Studio (VS) and PowerShell, allowing emojis and other special characters to display correctly.

1. Visual Studio Debugging Configuration (Recommended for VS Debugging)

This method modifies your project properties to launch your application within a PowerShell environment configured for UTF-8 when you run the **Debugger (F5)**.

1. Open Project Properties:

- In the **Solution Explorer**, **right-click** on your C++ project name.
- Select **Properties**.

2. Navigate to Debugging Settings:

- In the Property Pages window, go to **Configuration Properties → Debugging**.

3. Change Command and Command Arguments:

- Change the value for the **Command** field.
 - Set it to:
 - "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
- Change the **Command Arguments** field.
 - Set it to **Option A** (Recommended for standard debug execution):
 - -Command "& { [Console]::OutputEncoding = [System.Text.UTF8Encoding]::new(); Start-Process -FilePath '\$(TargetPath)' -NoNewWindow -Wait }"
 - *Alternatively*, use **Option B** (Simpler, but might not handle encoding as reliably):
 - -NoExit -Command "& \"\$(TargetPath)\""

2. Running from Windows Terminal/PowerShell (Manual Execution)

If you prefer to run the executable directly in a terminal, you need to manually configure the terminal's encoding.

Temporary Fix (Lost upon closing terminal)

1. **Open Terminal:** Launch **Windows Terminal** and select **PowerShell**.
2. **Navigate to Project Directory:**
3. `cd "C:\Users\qsami\source\repos\Wordle 2.1\x64\Debug"`

(Replace with your actual project path)

4. **Force Output to UTF-8:** Run this command *once* in the session:
5. `[Console]::OutputEncoding = [System.Text.UTF8Encoding]::new()`
6. **Run Code:** Execute your program:
7. `.\"Wordle 2.1.exe"`

Permanent Fix (Configures your PowerShell profile)

This changes the default encoding for every new PowerShell session.

STEP 1 — Check/Create a PowerShell Profile

- Open **Windows Terminal** → **PowerShell**, then run:
- `Test-Path $PROFILE`
- If it prints: `false`
 - Create it:
 - `New-Item -Path $PROFILE -Type File -Force`

STEP 2 — Open the PowerShell Profile

- Run the following command to open the profile file in Notepad:
- `notepad $PROFILE`
 - This will open a file path like:
`C:\Users\<you>\Documents\PowerShell\Microsoft.PowerShell_profile.ps1`

STEP 3 — Add UTF-8 Encoding to the Profile

- Paste this line into the file to force **UTF-8 output**:
- `[Console]::OutputEncoding = [System.Text.UTF8Encoding]::new()`
- **OPTIONAL — Also force UTF-8 input (Recommended)**: Add this line as well:
- `$OutputEncoding = [System.Text.UTF8Encoding]::new()`
- **Save → Close Notepad.**

Execution Policy Error Fix

If you encounter the error: "Microsoft.PowerShell_profile.ps1 cannot be loaded because running scripts is disabled on this system."

This means PowerShell's security settings prevent it from running the profile script. You have two options:

1. **Open PowerShell as Administrator** and re-try the profile creation/editing.
2. **Allow only the current user to run scripts (Recommended safe option):**
 - Run this command:
 - `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`
 - Press **Y** when asked.
 - This is generally safe because it affects only **your user**, and it only allows local scripts unless they are digitally signed.